

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in building and demonstrating production-ready AI systems - RAG pipelines, agentic LLM workflows, and real-time inference APIs. Proven ability to translate complex AI capabilities into tangible business outcomes: ~35% retrieval precision improvement in healthcare RAG systems, >30% hallucination reduction via grounding-based agentic workflows, and 60-70% reduction in manual review time through transformer-based automation. Strong technical communication skills honed across healthcare AI consulting, student leadership, and cross-functional stakeholder documentation. Hands-on Python, FastAPI, LangChain, Docker, and multimodal API developer - able to rapidly prototype, iterate, and demonstrate proof-of-concepts for diverse technical and non-technical audiences. Eager to apply AI-first engineering to voice and speech AI solutions in a customer-facing role.

CORE COMPETENCIES

Proof-of-Concept Development • API Integration & Demonstration • Technical Pre-Sales Support • RAG & Agentic AI Systems • LLM Workflows & Structured Outputs • Python / FastAPI / Flask • Docker & Cloud-Ready Deployment • Technical Storytelling & Documentation • LangChain / LlamaIndex • Multi-Provider AI API Integration • Customer-Facing Technical Communication • AI-First Workflow Design

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare AI & Consulting

- Architected and delivered **RAG systems** for clinical knowledge retrieval - recursive semantic chunking, transformer embeddings, vector search; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25% and reduced hallucinated outputs by >30%.
- Packaged ML/LLM inference as **FastAPI/Flask RESTful services**, containerized with Docker; translated complex AI capabilities into clean API surfaces for non-technical stakeholder demonstration, reducing post-deployment defects by ~30%.
- Built **predictive ML pipelines** (scikit-learn, XGBoost) for patient risk scoring; 15-20% recall improvement on high-risk categories - presented results to healthcare operations stakeholders with actionable recommendations.
- Maintained HIPAA-conscious data governance and authored stakeholder-facing system-limitation documentation - translating technical AI constraints into clear business guidance.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP & Video Analytics

- Built a **transformer-based video summarization system** across 5,000+ recorded sessions; reduced manual review time by **60-70%** (hours to <5 minutes per video) for non-technical staff users.
- Achieved ~85% **alignment** between AI-selected highlights and human-curated key moments; deployed as a **Flask API** with lightweight web interface for non-technical users - end-to-end from prototype to production.
- Implemented RAG-based document Q&A; reduced hallucinated outputs ~30% and raised contextual answer accuracy >85% in controlled tests.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- Optimized SQL/T-SQL stored procedures for Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built CI/CD pipelines with Jenkins; reduced deployment errors >30% and improved release cycle time ~35-40%.

SELECTED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation Shipped · Chrome Web Store

Full-stack browser extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider API integration with provider-specific payload handling, seven-day translation cache, and per-domain configuration for 29 sites - demonstrating rapid multi-vendor AI API integration and user-facing product delivery.

Manifest V3 · Content Scripts · Service Workers · Multi-Provider AI APIs · WebP/JPEG handling

Dream Decoder - AI-Powered Multimodal Creative Platform

Full-stack FastAPI + React/TypeScript app coordinating multimodal AI APIs (Perplexity Sonar + Replicate image models). Introduced **intermediate structured prompt transformation layers** improving contextual alignment ~30% and first-pass image success rate ~25-30% - a reusable pattern directly applicable to voice AI POC development.

FastAPI · React/TypeScript/Vite · Perplexity API · Replicate API · Structured Outputs

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake → Validation → Consistency Analysis → Fraud Scoring) with **schema-validated JSON contracts between agents**. RAG-grounded CPT/ICD validation against vector-indexed policy documents; explainable risk scoring with audit-ready traces - demonstrates complex technical solution design for domain-specific enterprise use cases.

LangChain · LlamaIndex · FastAPI · Vector Search · Structured Outputs · Agentic Orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL

AI/LLM: LangChain, LlamaIndex, RAG, Agentic Workflows, Structured Outputs, Prompt Engineering, Embeddings, Vector Search, Hugging Face Transformers

ML: scikit-learn, XGBoost, PyTorch, YOLOv8, OpenCV

AI APIs: OpenAI, Anthropic Claude, Perplexity, Replicate, Ollama - multi-provider integration

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, MLOps/LLMOps

Work Authorization: F-1 OPT - authorized to work in the US now; future sponsorship required